

Client Design Review

MILESTONE 1 · TSHIMO AGRI SUPPLIES NETWORK DESIGN

📍 Potchefstroom, Agriculture

📶 10.23.0.0/16

🛡️ Fault Isolation — Advanced

STUDENT

Maluleke, Kurhula Success

STUDENT NUMBER

48277444

PROJECT ID

CMPG325-2026-042

CLIENT ID

CLI-042

ASSIGNED CLIENT

Tshimo Agri Supplies

ASSIGNED CHALLENGE

Network Troubleshooting

1

Client Requirements

Stated brief requirements and the interpreted department structure



2

Physical Topology

Devices, roles, and cabling for the Potchefstroom HQ



3

Logical Topology

VLANs, subnets, gateways, and the routing model



4

IP Addressing Plan

The full 10.23.0.0/16 breakdown and why it satisfies both constraints



5

Initial GitHub Repository

Portfolio structure, README skeleton, and commit plan



DESIGN PHILOSOPHY

Every decision in this plan follows one rule: solve the requirement by **allocation**, not by re-architecture. Uniform subnet sizes, a single summarisable HQ route, and untouched reserved space mean the branch-office constraint and Change Request 1 are both already solved *before they happen*.

1.1 STATED REQUIREMENTS — FROM THE PROJECT BRIEF

The following requirements are given directly by the assignment brief and are not open to interpretation.

- ✓ **Assigned organisation:** Tshimo Agri Supplies (Potchefstroom) · Industry: Agriculture
- ✓ **Assigned addressing block:** 10.23.0.0/16 — must form the basis of the entire IP addressing plan
- ✓ Provide appropriate connectivity and network services for the assigned scenario
- ✓ **Design Constraint:** a branch office may open within 18 months — the addressing plan must allow for it without redesign
- ✓ **Change Request CR1:** the client signs 8 additional staff in one department — capacity must be added without redesigning the network
- ✓ Produce a working, testable Cisco Packet Tracer implementation and demonstrate the assigned challenge: **Network Troubleshooting (fault isolation scenario)**

1.2 INTERPRETED ORGANISATIONAL REQUIREMENTS — DESIGN ASSUMPTION

Tshimo Agri Supplies' department structure was defined as part of this design, based on a realistic staffing model for a small-to-medium agricultural supply distributor. It forms the basis for the VLAN and addressing plan in Sections 2–4.

DEPARTMENT	FUNCTION	CURRENT STAFF
VLAN 10 Administration / Management	HR, finance, general management	8
VLAN 20 Sales	Customer-facing sales & orders	12
VLAN 30 Warehouse / Logistics	Stock, dispatch, delivery — CR1 department	12 (+8 via CR1 = 20)
VLAN 40 IT / Server Room	DHCP, DNS and File/ERP services	3 servers

- i Why CR1 lands on Warehouse / Logistics**
CR1 is applied to Warehouse/Logistics because staffing growth in an agricultural supply business most plausibly occurs in stock handling and dispatch as client order volume grows — not in Admin or Sales, which scale far more slowly relative to warehouse throughput.

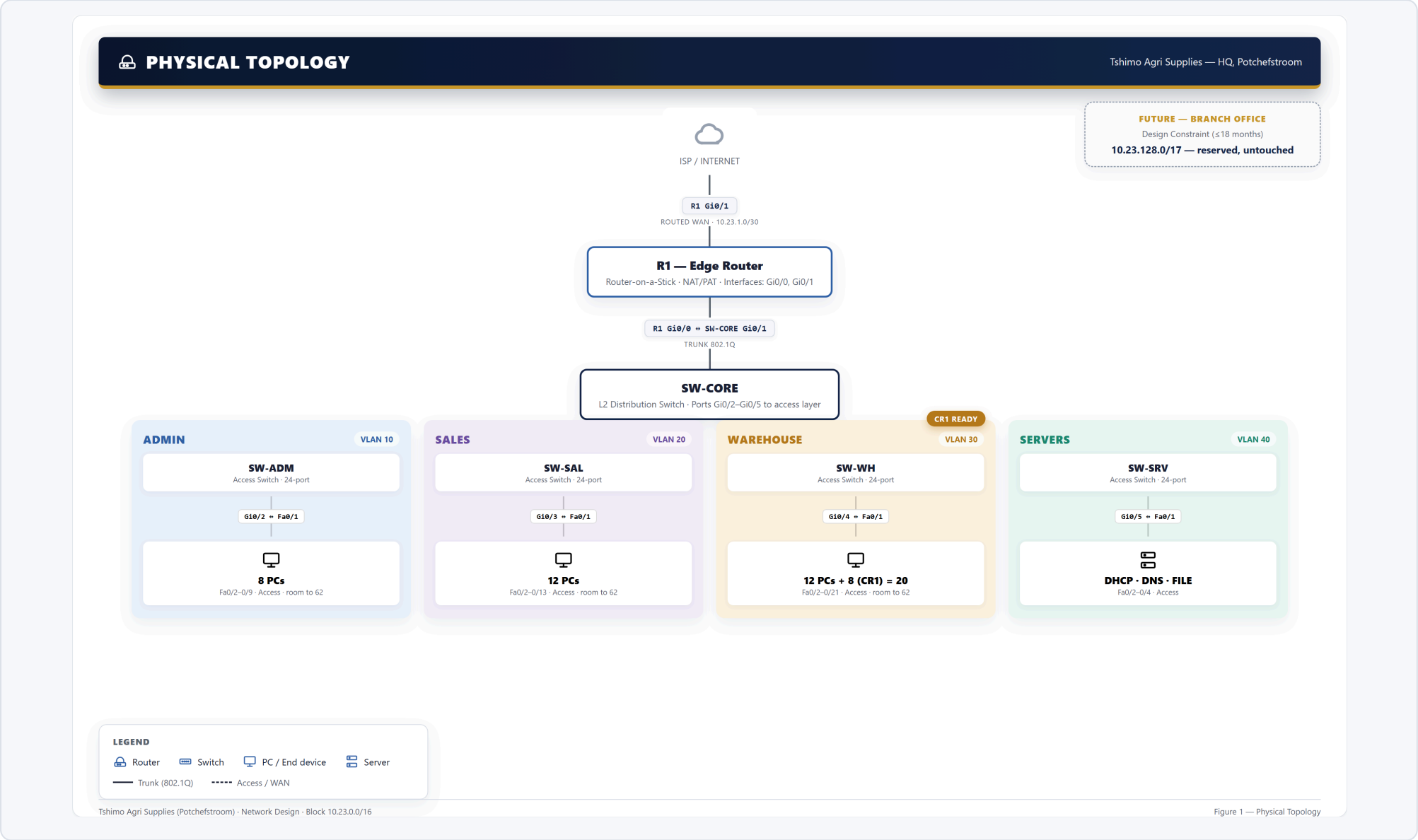


Figure 1 — Physical Topology, Tshimo Agri Supplies HQ (Potchefstroom). A single edge router (R1) provides the WAN connection; a core switch distributes 802.1Q trunks down to one 24-port access switch per department.

A single edge router provides the WAN connection; a core switch distributes 802.1Q trunks down to one access switch per department, keeping every department electrically and logically separate while sharing a single router for inter-department routing.

EDGE ROUTER

1 × R1 (Gi0/0, Gi0/1)

SWITCH LAYER

1 core + 4 access (24-port)

UPLINKS

802.1Q trunk, every link

WAN EDGE

NAT/PAT · 10.23.1.0/30

DEVICE	ROLE	SUGGESTED PT MODEL	KEY INTERFACES
 R1	Edge router — WAN uplink, inter-VLAN routing (router-on-a-stick), NAT/PAT	Cisco 4321 / 2911	Gi0/0 (trunk to SW-CORE), Gi0/1 (WAN to ISP)
 SW-CORE	Distribution switch — trunks VLANs between R1 and access layer	Cisco 2960	Gi0/1 (to R1), Gi0/2-Gi0/5 (to access switches)
 SW-ADM	Access switch — Admin/Management VLAN	Cisco 2960 (24-port)	Fa0/1 (uplink to SW-CORE Gi0/2), Fa0/2-0/9 (PCs)
 SW-SAL	Access switch — Sales VLAN	Cisco 2960 (24-port)	Fa0/1 (uplink to SW-CORE Gi0/3), Fa0/2-0/13 (PCs)
 SW-WH	Access switch — Warehouse/Logistics VLAN (CR1 department)	Cisco 2960 (24-port)	Fa0/1 (uplink to SW-CORE Gi0/4), Fa0/2-0/21 (PCs)
 SW-SRV	Access switch — server segment	Cisco 2960 (24-port)	Fa0/1 (uplink to SW-CORE Gi0/5), Fa0/2-0/4 (servers)
 End devices	Department PCs + 3 servers (DHCP, DNS, File)	PC-PT / Server-PT	FastEthernet to access switch, access mode



Physical headroom matches logical headroom

Access switches are specified as 24-port models even where current headcount is lower (e.g. Admin at 8 PCs), so CR1-style headcount growth is absorbed at the physical layer as well as the addressing layer — see Section 4.

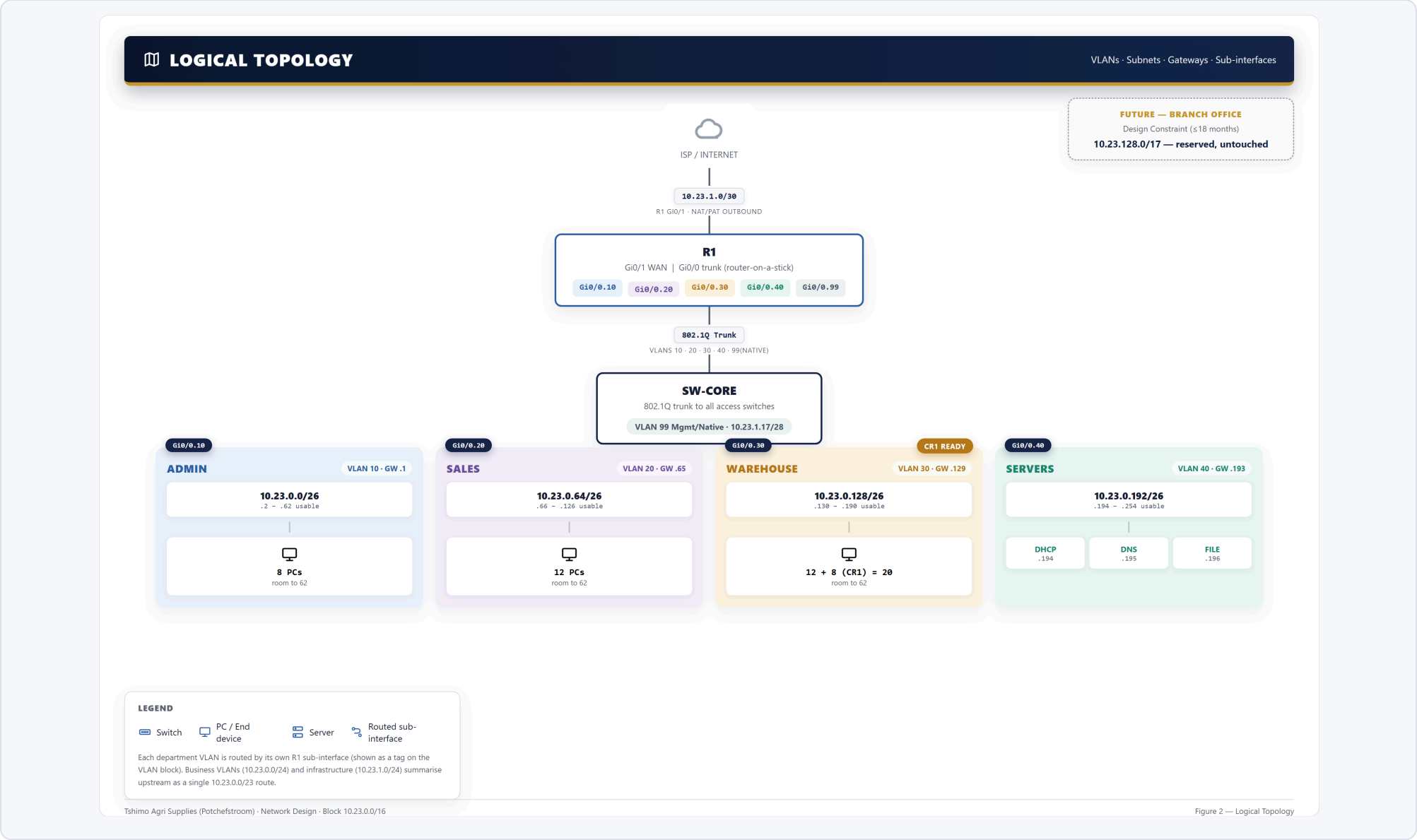


Figure 2 — Logical Topology: VLANs, subnets, gateways, and R1 sub-interfaces. R1 routes inter-VLAN traffic via router-on-a-stick (one sub-interface per VLAN on Gi0/0) and NATs internet-bound traffic out Gi0/1.

Each department is a separate VLAN. R1 performs inter-VLAN routing using a router-on-a-stick design (one sub-interface per VLAN on Gi0/0), and provides the default route out to the ISP on Gi0/1 with NAT/PAT for internet-bound traffic. DHCP is served from a dedicated DHCP server in the Server VLAN rather than the router, reflecting realistic enterprise practice.

VLANs

5 · 10/20/30/40/99

ROUTING

Router-on-a-stick, 1
sub-if / VLAN

DHCP

Dedicated server,
VLAN 40

SUMMARY ROUTE

10.23.0.0/23
upstream

VLAN	NAME	PURPOSE	SUBNET	R1 SUB-INTERFACE
10	ADMIN	Administration / Management	10.23.0.0/26	Gi0/0.10
20	SALES	Sales	10.23.0.64/26	Gi0/0.20
30	WAREHOUSE	Warehouse / Logistics (CR1 department)	10.23.0.128/26	Gi0/0.30
40	SERVERS	DHCP, DNS, File/ERP servers	10.23.0.192/26	Gi0/0.40
99	MGMT-NATIVE	Switch management / native VLAN (no user traffic)	10.23.1.16/28	Gi0/0.99



Why sub-interfaces, not a routed SVI per VLAN

Router-on-a-stick keeps all inter-VLAN routing logic on R1 alone, which matches the CMPG 325 syllabus and keeps the troubleshooting scenario's blast radius contained to a single device rather than spread across a Layer 3 switch and a router.

The assigned block **10.23.0.0/16** is split first at the /17 level: the lower half is allocated to the Potchefstroom HQ site now, and the upper half is reserved, untouched, for the branch-office constraint. Inside HQ, the plan uses exactly two /24 networks today — one for business VLANs, one for infrastructure — chosen so the whole HQ site collapses into a single summary route upstream.

WHY THIS IS THE SMART SOLUTION, NOT THE HARD ONE

- 1 Uniform block sizing**
 Every department VLAN is the same size ($/26 = 62$ hosts), so the whole plan fits in one sentence: "VLAN N lives at offset $(N/10 - 1) \times 64$ in 10.23.0.0/24." Nothing to memorise per department.
- 2 One summary route**
 10.23.0.0/24 (business) and 10.23.1.0/24 (infrastructure) are contiguous, so R1 can summarise the entire HQ site as a single 10.23.0.0/23 route upstream instead of advertising every subnet individually.
- 3 Separation of concerns**
 User traffic (business VLANs) and infrastructure (WAN, management, guest) live in different /24s — a real enterprise security practice, not just a numbering trick.
- 4 Growth by allocation, not redesign**
 CR1 and the branch-office constraint are both solved by handing out already-reserved space, never by re-cabling, re-VLANing, or re-IP-addressing anything already deployed.

4.1 TOP-LEVEL SPLIT — DESIGN CONSTRAINT (BRANCH OFFICE)

ALLOCATION	BLOCK	STATUS
HQ — Potchefstroom	10.23.0.0/17	In use — 2 of 128 possible /24s (see 4.2 / 4.3)
Reserved — future Branch Office	10.23.128.0/17	Untouched — available on demand



4.2 BUSINESS VLANS — 10.23.0.0/24 (UNIFORM /26 BLOCKS)

SUBNET (VLAN)	NETWORK / CIDR	USABLE RANGE	BROADCAST	GATEWAY	USABLE HOSTS
VLAN 10 Admin	10.23.0.0/26	.1 - .62	.63	.1	62 (8 used)
VLAN 20 Sales	10.23.0.64/26	.65 - .126	.127	.65	62 (12 used)
VLAN 30 Warehouse	10.23.0.128/26	.129 - .190	.191	.129	62 (20 used)
VLAN 40 Servers	10.23.0.192/26	.193 - .254	.255	.193	62 (3 used)

4.3 INFRASTRUCTURE — 10.23.1.0/24 (KEPT SEPARATE FROM USER VLANS)

PURPOSE	NETWORK / CIDR	USABLE RANGE	BROADCAST	GATEWAY	USABLE HOSTS
R1 – ISP WAN link	10.23.1.0/30	.1 - .2	.3	n/a	2 (point-to-point)
VLAN 99 Mgmt/Native	10.23.1.16/28	.17 - .30	.31	.17	14 (switches only)
Guest Wi-Fi (future)	10.23.1.32/28	.33 - .46	.47	.33	14 (reserved)
Free / future infra	10.23.1.48 - .255	—	—	—	208 addresses spare

4.4 HOW THIS PLAN SATISFIES THE CONSTRAINT AND CR1

DESIGN CONSTRAINT

Branch Office

The entire 10.23.128.0/17 (32,768 addresses) is reserved and untouched. Opening a branch requires only allocating from this pool — no re-addressing of the HQ site.

CHANGE REQUEST CR1

+8 Staff, No Redesign

Warehouse/Logistics (VLAN 30) is sized at /26 (62 usable) against 12 current staff. Adding 8 brings usage to 20/62 — well inside the existing subnet. No VLSM change, no re-cabling.

General headroom: 126 further /24 networks remain free inside the HQ /17 alone, before any branch-office space is touched — this design will not need revisiting for the foreseeable life of the business.

Repository: **tshimo-agri-network** — github.com/MALULEKE-KS/tshimo-agri-network — the individual GitHub portfolio required by the module brief, tracking design documentation, configuration, and evidence through to final submission.

5.1 FOLDER STRUCTURE

```
tshimo-agri-network/  
  README.md — overview, client summary, milestone tracker  
  docs/ — living design documentation  
    01-client-requirements.md  
    02-physical-topology.md  
    03-logical-topology.md  
    04-ip-addressing-plan.md  
    05-change-requests.md  
  assets/ (physical/logical topology PNGs, banner)  
  submissions/ — frozen, graded deliverables per milestone  
    milestone-1/ (this PDF, .docx source, eFundi text)  
  packet-tracer/ — final .pkt file (later milestones)  
  configs/ — per-device Cisco IOS command sets  
  screenshots/ — connectivity and configuration evidence  
  troubleshooting/ — fault isolation documentation (assigned challenge)
```

5.3 INITIAL COMMIT SEQUENCE

- 1 `git init` in project folder, add README.md and /docs skeleton
- 2 Commit 1 — add client requirements and project brief summary
- 3 Commit 2 — add physical and logical topology diagrams
- 4 Commit 3 — add IP addressing / VLSM plan
- 5 Push to GitHub and paste the repository link into the eFundi text box alongside this document

PROJECT TIMELINE

☒	Commencement	14 Aug 2026	Complete
☒	Milestone 1 — Design Review	28 Aug 2026	Package prepared
☐	Milestone 2	2 Oct 2026	Not started
☐	Final Submission	16 Oct 2026	Not started

DECLARATION

I declare that this design and its accompanying documentation are my own work, produced for CMPG 325 at North-West University, Mahikeng campus, in accordance with the NWU AI Policy. All design decisions above — department structure assumptions, VLAN/subnet allocation, and topology choices — are ones I understand and can defend.

Signature — Kurhula Success Maluleke

Date